Towards Clinically Trustworthy AI for Pancreatic Cancer Diagnosis

This document provides a concise overview of my master's thesis project, which investigates the use of machine learning and explainable artificial intelligence (XAI) to support the detection of pancreatic cancer. The project explores how predictive models built on clinical data can be complemented by explainability techniques to make their outputs more transparent, interpretable, and suitable for clinical contexts.

Beyond technical performance, the study places particular emphasis on the usability and trustworthiness of AI explanations from the perspective of healthcare professionals. To address this, the project includes a qualitative evaluation phase involving domain experts.

This summary is intended to present the project in a clear and operational way, highlight potential areas of collaboration, and outline how clinical and research expertise can contribute to ensuring that the proposed approach aligns with real-world medical needs.

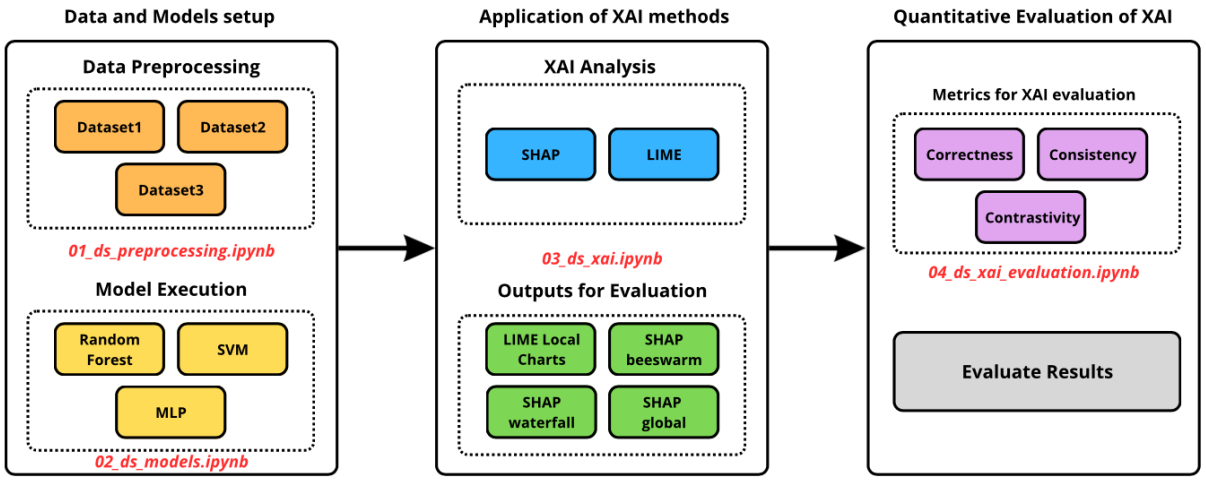
1. **Research Motivation**

Pancreatic Ductal Adenocarcinoma (PDAC) remains one of the most lethal malignancies, largely due to late-stage diagnosis and the absence of specific early symptoms. In recent years, machine learning models applied to clinical biomarkers and imaging data have shown promising results in supporting early detection and classification tasks. However, despite encouraging performance metrics, the clinical adoption of these systems remains limited. The main barrier is not only accuracy, but transparency: most predictive models operate as black boxes, providing outputs without clear justification.

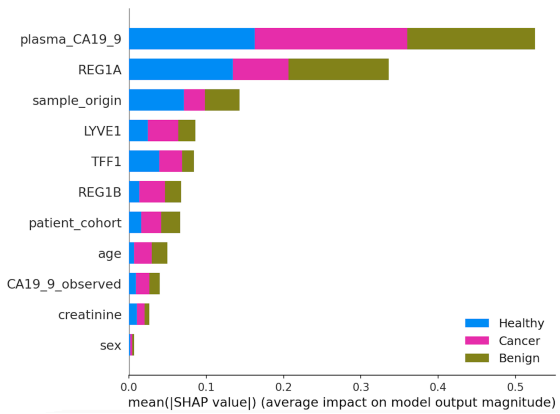
In high-stakes contexts such as oncology, where diagnostic decisions directly impact treatment pathways and patient survival, a prediction alone is not sufficient. While Explainable AI (XAI) techniques aim to make model decisions more interpretable, there is still a significant gap in understanding whether these explanations are truly usable, trustworthy, and clinically meaningful from the perspective of healthcare professionals. Systematic evaluations of the clinical usability of AI explanations are still scarce, and this gap motivates the present work.

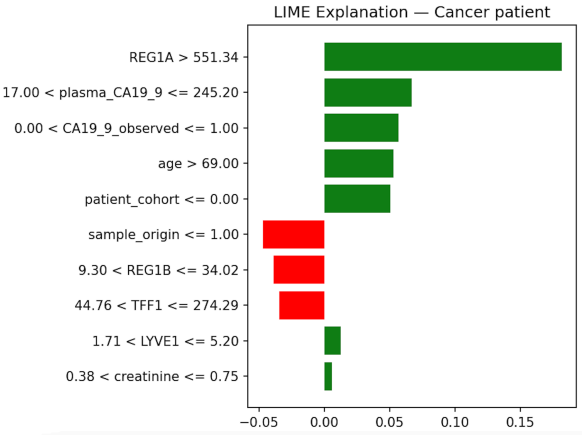
2. Technical Implementation

The crucial part of the thesis will focus on the processing of datasets, implementation and evaluation of models, XAI analysis and their related mixed evaluation. Below you can find the technical pipeline that explains the evolution of the project stored in the Github repository:



STEP 1: DATA PREPARATION
STEP 2: MODELS EVALUATION - Random Forest, Support Vector Machine, MLP Classifier
STEP 3: XAI Analysis with SHAP and LIME. Sample outputs of the XAI analysis:





STEP 4: XAI Evaluation via:

a. **Quantitative Analysis**: definition of quantitative metrics that check the robustness, reproducibility and consistency of the explainable metrics
b. **Qualitative evaluation:** domain experts will evaluate SHAP and LIME explanations via a structured survey assessing understandability, clinical coherence, sufficiency, clarity, and trust. Results will be mapped to IEEE P7001 transparency levels (0–5) to assess clinically meaningful transparency.

The final evaluation of both XAI techniques, that are LIME and SHAP, will consist not only of the quantitative parts as it happens in most research works, but will also comprise a qualitative evaluation that takes into account also domain experts' perspective.

3. Potential Collaboartion

Your expertise would be highly valuable in ensuring that the explainability outputs are clinically meaningful and aligned with real-world decision-making processes.

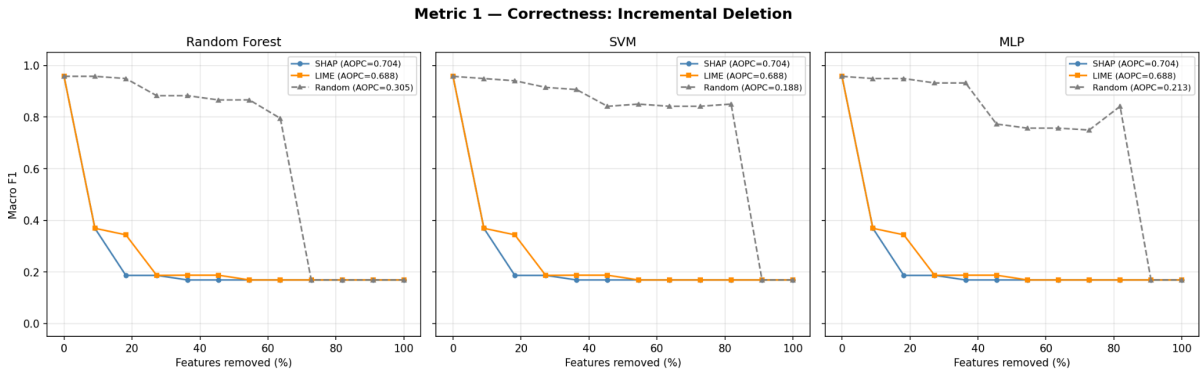
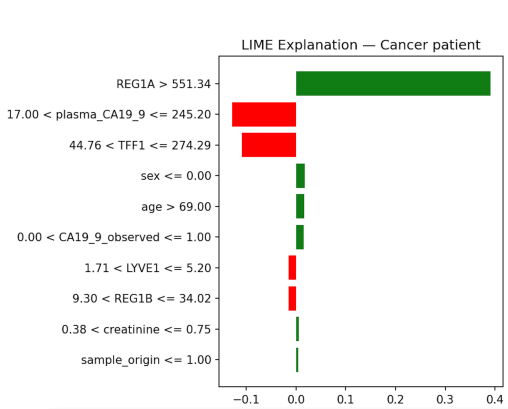
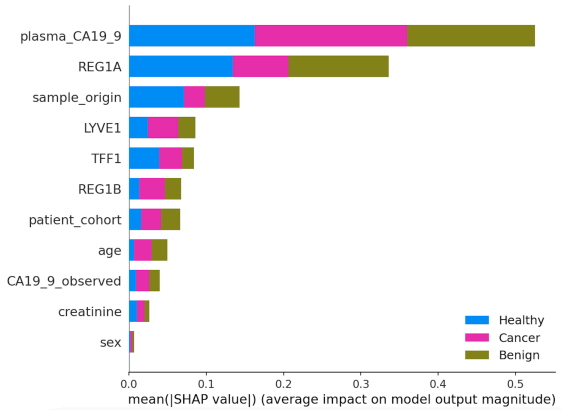
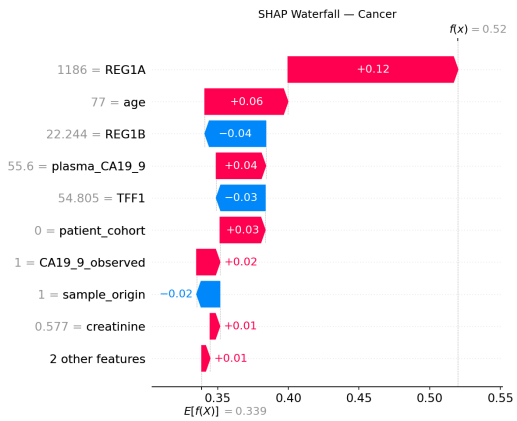
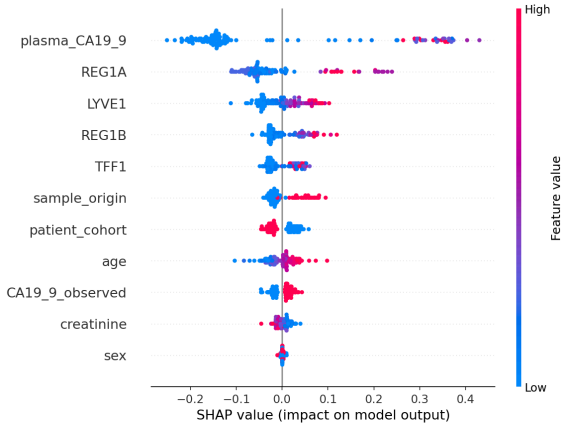
Specifically, your contribution may include:

1. **Qualitative evaluation of XAI outputs**
   Reviewing selected explanation outputs (e.g., feature rankings and image-based highlights) and providing feedback through a short structured survey and/or brief interview. The evaluation will focus on interpretability, clinical coherence, usefulness, and trust.
2. **Clinical perspective on interpretability needs**
   Sharing insights on what makes AI explanations understandable and actionable in clinical practice, and which types of explanations are most relevant for supporting diagnostic reasoning.
3. **Optional: involvement of colleagues or students**
   If appropriate, involving additional clinicians, researchers, or medical students to broaden the evaluation and capture diverse perspectives.
4. **Optional: data collaboration**
   If datasets relevant to pancreatic cancer research are available, their use in anonymized form or under NDA could significantly enhance the clinical relevance of the study. Any data use would fully comply with ethical and legal requirements, and no sensitive data would be published.

Appendix

Github of the code repository:
https://github.com/ariannacalderara/Pancreatic_Cancer_Detection_Master_Thesis

Example of quantitative analysis on XAI methods:



Graphics that will have to evaluated via surveys/interviews: